

NEUROLXP

Attendance Management Flow-wise Module Documentation

Simple functional documentation for University, Skilling Academy, Bootcamp and Corporate tenants

1. What We Are Building

Attendance Management is one common LMS feature used by different tenant types. The screens and engine remain common, while names, learning structure, attendance rules and methods can change tenant-wise.

Simple product idea

Start with who is learning and what they are enrolled in. Create the learning calendar and sessions. Take attendance. Review it. Calculate the result. Then show alerts, actions and reports.

2. One Common Flow for Every Tenant

TENANT → STRUCTURE → COURSE/TRAINING → CLASS/BATCH/COHORT → ENROLLMENT → CALENDAR → SESSION → ATTENDANCE → REVIEW → RESULT → ACTION → REPORT

The same flow is reused for every tenant. Only the business names and attendance rules change.

3. Tenant-wise Meaning

Tenant	Learning structure	Attendance is mainly for	Person taking/reviewing attendance
University	Department → Program → Semester → Section → Subject	Scheduled class/lab	Faculty
Skilling Academy	Center → Course → Batch	Daily training session	Trainer
Bootcamp	Program → Cohort → Module	Live session/workshop	Instructor/Mentor
Corporate	Business Unit/Team → Training Program	Employee training/workshop	Trainer/L&D or automatic integration

4. Roles and Responsibilities

Role	Simple responsibility
Super Admin	Controls the LMS platform and tenant availability.
Platform Admin	Onboards/supports tenants and enables Attendance capabilities.

Institute/Tenant Admin	Creates tenant learning structure, enrollment setup, calendar, policies and reports.
Department/Center/L&D Admin	Manages attendance for the allowed department, center or team scope.
Faculty/Trainer/Instructor	Takes and reviews day-to-day attendance.
Learner/Student/Employee	Checks in when allowed, views own attendance and raises permitted requests.

5. Flow 1 - Tenant and Learning Structure Setup

CREATE TENANT → CHOOSE TENANT TYPE → CREATE ORGANIZATION STRUCTURE → CREATE LEARNING OFFERING → CREATE LEARNING GROUP

Examples:

Tenant	Example setup
University	Adichunchanagiri University → Information Science → B.E/B.Tech → Semester 3 → Section A → Data Structures
Skilling Academy	SkillPro Academy → Bengaluru Center → Full Stack Course → Batch FSD-01
Bootcamp	CodePro → Full Stack Bootcamp → September Cohort → React Module
Corporate	ABC Technologies → Engineering → Development Team → Cybersecurity Training

6. Flow 2 - Enrollment

CREATE/IMPORT USERS → SELECT COURSE/GROUP → ENROLL LEARNERS → ASSIGN FACULTY/TRAINER → CREATE ACTIVE ROSTER

Enrollment decides who is expected to attend. A learner should appear in attendance only when the learner is actively enrolled in the correct course, subject, batch, cohort or training.

- University: enroll students into program/semester/section/subjects.
- Academy: enroll learners into course and batch.
- Bootcamp: enroll learners into program and cohort.
- Corporate: assign employees to training, audience or mandatory program.

7. Flow 3 - Calendar and Timetable

CREATE CALENDAR → ADD TERM/BATCH/TRAINING DATES → ADD HOLIDAYS → CREATE TIMETABLE/SCHEDULE → ASSIGN FACULTY + LOCATION/LINK

The calendar tells the LMS when learning should happen. The timetable or schedule tells it the exact recurring or one-time class/session.

Item	Example
Calendar	Semester 3: Aug-Dec 2026
Holiday	University holiday - no class
Schedule	Data Structures - Mon/Wed/Fri - 10:00 to 11:00

Faculty	Prof. Kumar
Location	Room 204 or virtual meeting link

8. Flow 4 - Attendance Session Creation

TIMETABLE/SCHEDULE → LMS CREATES SESSION → LOAD ELIGIBLE LEARNERS → APPLY ATTENDANCE POLICY → UPCOMING SESSION

Faculty should not create a fresh attendance sheet from zero every day. The LMS should normally generate the session from the schedule and bring the correct learner roster.

Session status	Meaning
Upcoming	Future session.
Open	Attendance can be recorded.
Review	Check-in is closed and faculty reviews exceptions.
Submitted	Faculty has submitted.
Finalized	Official attendance result.
Cancelled	Session does not count.

9. Flow 5 - Attendance Policy

TENANT DEFAULT POLICY → OPTIONAL COURSE/GROUP OVERRIDE → SESSION USES FINAL POLICY

- Allowed attendance method: Manual, QR, Dynamic QR, GPS, virtual integration, etc.
- When attendance opens and closes.
- Grace period and Late rule.
- Minimum session duration if required.
- Minimum overall attendance percentage.
- How Excused, Late and Partial statuses are treated.
- Who can edit/correct attendance and for how long.

10. Flow 6 - Types of Attendance

Type	Simple meaning	Good for
Manual	Faculty marks each learner or uses bulk marking.	All tenants
Dynamic QR	Learner scans a short-lived/rotating QR.	University, Academy, Bootcamp
PIN/Code	Learner enters a temporary session code.	Backup/Hybrid
GPS/Geofence	Check-in is allowed near an approved place.	Campus/Training center
Virtual attendance	Join/leave information is used.	Corporate, Bootcamp, Online University
Activity-based	Required learning activity becomes attendance evidence.	Async/Bootcamp
Offline attendance	Faculty records without internet and syncs later.	Academy/Campus
External attendance	Approved access/RFID/other system sends attendance evidence.	Optional enterprise/campus

11. Flow 7 - Faculty/Trainer Attendance Management

LOGIN → TODAY'S SESSIONS → OPEN SESSION → TAKE ATTENDANCE → LIVE ROSTER → REVIEW → SUBMIT

- Faculty sees only assigned sessions.
- Faculty uses only methods allowed by the tenant policy.
- Live roster shows Present, Late, Not Checked In, Absent/Pending as applicable.
- Faculty can correct permitted mistakes before submission.
- After submission/finalization, later changes use a controlled correction flow.

12. Flow 8 - Learner Attendance

**LOGIN → TODAY'S LEARNING → OPEN SESSION → CHECK-IN/JOIN OR FACULTY MARKS → CONFIRMATION
→ MY ATTENDANCE**

- If self check-in is enabled, learner scans QR/uses approved method.
- System checks the learner, session and allowed time.
- Learner receives a clear success, Late, Pending or rejected message.
- After finalization, learner sees the official status and updated percentage.

13. Flow 9 - Attendance Processing

ATTENDANCE INPUT → VALIDATE → APPLY RULES → PROVISIONAL STATUS → FACULTY REVIEW → FINAL STATUS

Situation	Possible result
Learner checks in within allowed time	Present
Learner checks in after grace period	Late
No valid attendance before closing	Absent
Learner attends only part of required online duration	Partial
Approved leave/valid exception	Excused
Information needs staff review	Pending

14. Flow 10 - Review, Corrections and Leave

ISSUE/REQUEST → FACULTY OR ADMIN REVIEW → APPROVE/REJECT → UPDATE STATUS → RECALCULATE → NOTIFY

- Attendance correction: learner says attendance was recorded incorrectly.
- Leave/excuse: learner submits a reason and, where required, supporting document.
- Late external attendance: virtual attendance information may arrive after the class.
- Cancelled/rescheduled class: the session must not create an incorrect absence.
- Every approved change should remain visible in the attendance history/audit.

15. Flow 11 - Results

FINAL ATTENDANCE → CALCULATE % / DURATION / COMPLIANCE → APPLY THRESHOLD → SHOW STATUS

Tenant	Typical final result
University	Subject/class attendance %, shortage status, semester/department view.
Skilling Academy	Course/batch attendance %, consistency and make-up needs.
Bootcamp	Session/cohort attendance plus optional duration/engagement view.
Corporate	Training attendance/compliance and mandatory-training status.

Important

Each tenant must decide how Present, Late, Partial and Excused affect the final percentage. The LMS should allow the tenant to configure this.

16. Flow 12 - Alerts and Actions

RESULT CHANGES → CHECK RULES → CREATE ALERT/ACTION → SEND TO CORRECT ROLE

- Successful check-in confirmation.
- Late or absence notification.
- Warning before attendance falls below the required percentage.
- At-risk escalation to faculty/HOD/manager/admin.
- Correction/leave decision notification.
- Make-up session action for Academy/Bootcamp where enabled.
- Mandatory-training follow-up for Corporate.
- Certificate/completion eligibility action where the tenant uses attendance as a requirement.

17. Flow 13 - Dashboards and Reports

User	What they need
Institute/Tenant Admin	Overall attendance, department/center comparisons, at-risk learners, sessions, trends and reports.
HOD/Center/L&D Admin	Only their allowed scope, low attendance and unresolved cases.
Faculty/Trainer	Today's sessions, class roster, attendance %, missing learners and requests.
Manager	Assigned employee training attendance/compliance.
Learner	Own attendance %, subject/course history, alerts and requests.

18. Flow 14 - AI in Attendance

ATTENDANCE HISTORY → AI ANALYSIS → INSIGHT/RECOMMENDATION → HUMAN OR RULE-BASED ACTION

AI use	Benefit	Limitation / caution
Risk prediction	Warn before a learner drops below threshold.	Prediction can be wrong; staff should review important actions.
Trend insights	Find repeated absence/late patterns.	Bad or incomplete data can create

		misleading insights.
Admin summaries	Quickly explains large attendance trends.	Summary must not replace underlying report.
What-if guidance	Shows how future attendance may affect percentage.	Use exact deterministic calculation for official numbers.
Anomaly assistance	Flags unusual attendance behavior for review.	Do not automatically accuse or punish a learner.

AI rule

AI should assist with insights, prioritization and recommendations. Official attendance should come from configured attendance rules and authorized review, not from an AI guess.

19. Additional Features That Make Attendance Attractive

- Dynamic QR and optional geofencing.
- Attendance risk alerts before threshold breach.
- Attendance what-if calculator.
- Heatmaps and trend insights.
- Leave and correction workflows.
- Make-up sessions.
- Offline faculty attendance.
- Bulk attendance with exception editing.
- Virtual meeting attendance.
- Optional attendance streaks/goals.
- Attendance + engagement view for Bootcamp/Corporate.
- Multi-campus/multi-center support.
- Configurable notifications and escalation.

20. Complete Flow by Tenant

20.1 University

UNIVERSITY → DEPARTMENT → PROGRAM → SEMESTER/SECTION → SUBJECT → ENROLL STUDENTS → TIMETABLE → CLASS SESSION → FACULTY ATTENDANCE → SUBJECT % → SHORTAGE ACTION

Example: Adichunchanagiri University → Information Science → Semester 3 → Section A → Data Structures.

20.2 Skilling Academy

ACADEMY/CENTER → COURSE → BATCH → ENROLL LEARNERS → BATCH CALENDAR → TRAINING SESSION → TRAINER/QR/GPS → BATCH % → MAKE-UP/FOLLOW-UP

20.3 Bootcamp

PROGRAM → COHORT → MODULE → LIVE/WORKSHOP SESSION → JOIN/CHECK-IN + DURATION → REVIEW → ATTENDANCE + OPTIONAL ENGAGEMENT → INTERVENTION

20.4 Corporate

**ORGANIZATION → TEAM/AUDIENCE → TRAINING → ASSIGN EMPLOYEES → TRAINING CALENDAR →
SESSION → VIRTUAL/MANUAL ATTENDANCE → COMPLIANCE → MANAGER/L&D ACTION**

21. Final End-to-End Product Flow

1 TENANT ONBOARDING

↓ **2 STRUCTURE / DEPARTMENT / CENTER / TEAM**

↓ **3 COURSE / SUBJECT / PROGRAM / TRAINING**

↓ **4 SECTION / BATCH / COHORT / AUDIENCE**

↓ **5 ENROLLMENT + FACULTY/TRAINER ASSIGNMENT**

↓ **6 CALENDAR + HOLIDAYS + TIMETABLE/SCHEDULE**

↓ **7 LMS CREATES SESSION + ROSTER**

↓ **8 ATTENDANCE OPENS**

↓ **9 MANUAL / QR / GPS / VIRTUAL / OTHER APPROVED METHOD**

↓ **10 VALIDATE + APPLY TENANT RULES**

↓ **11 FACULTY REVIEW + SUBMIT + FINALIZE**

↓ **12 CALCULATE ATTENDANCE RESULT**

↓ **13 ALERTS / CORRECTIONS / LEAVE / MAKE-UP / COMPLIANCE**

↓ **14 DASHBOARDS + REPORTS + OPTIONAL AI INSIGHTS**

Final product output

A clear, tenant-specific attendance result for every eligible learner and session, plus learner history, attendance percentage/compliance, staff actions, reports and audit history.